

Emmanuel A. Ogunniyi

Curriculum Vitae

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Professional Academic Experience

- November 2025 - present **Research Assistant Professor**, School of Science and Engineering, Benedict College, Columbia, South Carolina.
- October 2025 - present **Research Associate**, Department of Mechanical Engineering, University of South Carolina, Columbia, South Carolina.
- January 2021 - August 2025 **Graduate Research Assistant**, Major advisor: Austin Downey, Department of Mechanical Engineering, University of South Carolina, Columbia, South Carolina.
- February 2020 - December 2020 **Research Scholar**, Department of Mechanical Engineering, Jomo Kenyatta University of Agriculture and Technology, Kiambu, Kenya.
- January 2018 - November 2018 **Undergraduate Research Assistant**, Major advisor: Olurotimi Dahunsi, Department of Mechanical Engineering, Federal University of Technology, Akure, Ondo State, Nigeria.

Education

- 2025 **Ph.D**, Mechanical Engineering, University of South Carolina, Columbia, South Carolina. Dissertation: “Reduced-Order Model-Based Framework for Microsecond Model Updating of Two-Dimensional Structural Systems Using the Local Eigenvalue Modification Procedure”
- 2023 **MSc**, Mechanical Engineering, University of South Carolina, Columbia, South Carolina. GPA 4.0/4.0. Thesis: “Elastic Sensing Skin for Monitoring Concrete Structures”
- 2018 **B.Sc.**, Mechanical Engineering, Federal University of Technology, Akure. GPA 3.75/4.0. Senior project: “Integration of Pantograph Mechanism in the Design and Development of an Automatic Template Tracing Machine”

Research Highlights

- In Situ Manufacturing Monitoring** In situ monitoring and online validation of additively manufactured components using a variety of sensing systems focusing on real-time quality validation of components during the manufacturing process. Applications include fused-filament fabrication (FFF), metal additive manufacturing etc.
- Smart Materials and Innovative Sensing** The research in this field involves the development of smart materials and innovative sensing technologies for enhanced structural health monitoring. Key contributions include the creation and integration of large area capacitive sensors, and nanocomposite sensors for damage detection in diverse structures such as concrete infrastructures and composite materials etc.
- Digital Twins and smart Manufacturing** This research spans digital twin-enabled smart manufacturing systems integrating robotics, automation, and control for intelligent production. The work focuses on AI-enhanced process optimization, real-time cyber-physical synchronization, and secure digital manufacturing systems for automated and resilient manufacturing environments.

High-Rate Machine Learning This research area combines advanced structural modeling with high-rate machine learning, focusing on real-time decision making for dynamic environments. Key contributions include the development of computational models for complex structural analysis and high-speed machine learning algorithms for rapid data processing and anomaly detection in high-stress scenarios.

Battery safety and management systems Explores advanced management techniques for enhancing battery safety and efficiency, focusing on real-time monitoring and control. Applications span electric vehicles, renewable energy storage, and portable electronic devices. The goal is to enhance the reliability, longevity, and performance of battery systems while mitigating risks of failure and improving overall energy efficiency.

Curriculum Vitae Statistics

Publications Over 55 citations, h-index 5, 15+ publications
Students Actively mentored over 15 undergraduate students with significant research
Funding Received over \$300K in funding towards doctoral degree
Funding Actively participated in research funding worth over \$10M

Online Profile

ORCID <https://orcid.org/0000-0001-5456-5929>
LinkedIn <https://www.linkedin.com/in/eaogunniyi/>
Google Scholar <https://scholar.google.com/citations?user=9c7QBxEAAAAJ&hl=en&oi=ao>
ResearchGate <https://www.researchgate.net/profile/Emmanuel-Ogunniyi-2>

Research Experience

November 2025 - Present **Research Assistant Professor**, School of Science and Engineering, Benedict College, Columbia, South Carolina.

- Leading the Continuous Fiber Printing (CFP) research thrust, including design, setup, validation, and data analytics.
- Developing and implementing sensing and monitoring systems for CFP, integrating optical, thermal, and mechanical sensing for in-situ process validation.
- Conducting machine learning-based model updating and reduced-order modeling to enable real-time defect detection and process optimization.
- Mentoring undergraduate and master's students in research methods, additive manufacturing, and sensing-based data analysis.
- Prepare quarterly and annual project reports, documenting research outcomes, student progress, and experimental findings.
- Preparing manuscripts, proposals, and conference presentations derived from the project's outcomes.
- Supporting the establishment and maintenance of the additive manufacturing laboratory, including equipment operation, safety, and procurement.

January 2021 - **Graduate Research Assistant**, Adaptive Real-Time Systems Laboratory, USC-
September 2025 Columbia, South Carolina.

Collaborated on interdisciplinary research in advanced manufacturing, materials science, and smart systems, contributing to federal and industry-sponsored projects. Designed, fabricated and tested smart structures and additive manufactured materials, improving performance, cost-efficiency, and sustainability. Assisted in grant writing, equipment acquisition, and streamline laboratory operations to support multiple concurrent research initiatives

- Conducted structural (3D CAD) modeling and simulation, employing finite element analysis (FEA) to assess parameters such as yield strength, stress, strain, and deformation across critical components using ANSYS, SolidWorks, and ABAQUS, achieving over 95% simulation accuracy.
- Improved 3D-printed part reliability by testing for defects using dynamic testing systems and digital image correlation, achieving a 99.8% defect detection rate, which significantly lowered post-processing costs.
- Conducted rigorous impact, shock, and vibration testing to identify failure modes in electronic components, leading to targeted design improvements and a 15% reduction in defect rates, enhancing product reliability and lifecycle.
- Developed a reduced-order model in C and Python for real-time state estimation of dynamic systems, achieving computation speeds $125\times$ faster and maintaining error rates below 10%, significantly outperforming traditional solvers.
- Resolved long-standing capacitive coupling issues in flexible sensors by redesigning sensor architecture and applying advanced signal conditioning, achieving over 99% data accuracy and surpassing commercial alternatives.
- Engineered a multifunctional structural battery by embedding LiPo cells and flexible sensors in fiberglass, enabling real-time impact monitoring with more than 95% accuracy, reducing system weight by 20%, and eliminating separate sensor layers.
- Conducted failure analysis of materials using advanced characterization methods, including CT scanners, OMM, CMM, Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), Energy Dispersive Spectroscopy (EDS), and FTIR, to assess internal defects, microstructural features, and material composition.
- Led the design of automated drop tower test setups, integrating NI PXI hardware, BK precision LCR meter, and LabVIEW-based control systems, enabling 95% repeatability in high-speed impact simulations and cutting setup time by 50%

January 2021 - **Research Assistant**, Department of Transportation, Pooled fund, Columbia, SC.
May 2024

Collaborated with cross-functional engineering teams to translate simulation results and field data into actionable insights to monitor infrastructure health and system design improvements

- Compiled and delivered 30+ detailed technical reports on sensor performance and field tests, supporting federal and academic reviews that secured ongoing project funding and facilitated publications.

- Validated capacitive sensor performance on concrete structures under dynamic loading, achieving accuracy within $\pm 5\%$ of benchmarks, improving the reliability of structural health monitoring, and supporting proactive maintenance decision.
- Deployed flexible sensor arrays for high-resolution fatigue crack monitoring on bridges, achieving under 10% error in detection and localization, enabling early maintenance interventions, and reducing unplanned repairs.

February 2020 - **Pan African Scholar**, Department of Mechanical Engineering, Jomo Kenyatta University of Agriculture and Technology, Kiambu.

- Developed a compact, electrically insulated shower head with integrated heating and safety sensors, reducing temperature fluctuations by 85% and eliminating electrocution risks, meeting all electrical safety standards.
- Designed a modular, rechargeable electric iron using sustainable engineering principles, reducing production complexity and material waste by 25% and boosting battery life efficiency by 30%.

January - **Undergraduate Research Assistant**, Department of Mechanical Engineering, Federal University of Technology, Akure, Ondo State, Nigeria.

- Utilized a pantograph mechanism to innovate a robotic arm capable of performing tasks such as writing, drawing, and tracing templates.
- Designed a solar-powered pump and conducted research to analyze the correlation between the flow rate and the head of the solar pump.
- Led a research project on "Smart Home Technologies" aimed at optimizing energy consumption within homes to significantly save and extend the power supply in Akure, Ondo State, achieving a 20% reduction in household energy use through innovative automation and IoT solutions.
- Developed and conducted a performance evaluation of a quadcopter, focusing on its aerodynamic efficiency, which led to significant improvements in flight stability, maneuverability, and energy consumption.

Teaching Experience

Spring 2022 - **Graduate Teaching Assistant**, University of South Carolina, Columbia, SC.
Fall 2023

- Evaluated and graded assignments, exams, and coursework for undergraduate engineering classes, ensuring timely and constructive feedback to support student learning and performance improvement.
- Delivered hands-on laboratory instruction for Mechatronics courses, guiding students through practical experiments and applications of control systems, sensors, and actuators in real-world engineering problems.
- Provided mentorship to undergraduate and master's students, offering academic guidance, research supervision, and career development support to foster growth and professional advancement.

March 2019 - **Graduate Teaching Assistant**, Kogi State Polytechnic, Lokoja, Nigeria.
February 2020

- Conducted lectures and managed project assessments for undergraduate mechanical engineering design course.
- Monitored various class and student metrics, which encompassed attendance records, project progress, laboratory performance, and report submissions.

Industrial Experience

June 2017 - **Mechanical Engineer Intern**, Shongai Technologies Limited, Ogun State, Nigeria.
December 2017

- Overhauled maintenance procedures for critical manufacturing equipment, including the Blown Film Laminate Machine, Flexo-Printing Machine, Super Simplex Lamination Machine, and Slitting Machine, directly resulting in a 30% reduction in downtime and smoother production flows.
- Carried out data collection, root cause analysis (5-whys), corrective actions, and Failure Mode and Effect Analysis (FMEA).
- Implemented 5S and 8D problem-solving techniques, improving the manufacturing process and product quality by over 70%, demonstrating a strong understanding of manufacturing efficiency and waste reduction..
- Maintained process documentation including process flowcharts, PFMEA, control plans, process sheets, visual aids, gage instructions, control charts, and capacity to improve factory quality.

November 2010 - **Technician**, Johnson & Co Engineering Ltd, Ogun State, Nigeria.
December 2011

- Proficient execution of engineering tasks such as welding, drilling, grinding, and lathe machining.
- Implemented a comprehensive maintenance protocol for workshop machines and equipment, significantly reducing downtime and extending the lifespan of tools.

Journal Publications

- [9] **Emmanuel A. Ogunniyi**, Austin RJ Downey, and Yanhzou Fu. Multi-sensor data fusion for in-situ monitoring and quality control in additive manufacturing: A review. *Preprint in review for submission*
- [8] **Emmanuel A. Ogunniyi**, Austin RJ Downey, Pete Avitabile, Jacob Dodson, and Adraine G Moura. $O(n)$ digital twin updating of finite element structural models via the local eigenvalue modification procedure with microsecond-latency performance. *Preprint to be submitted to Mechanical Systems and Signal Processing*
- [7] **Emmanuel A. Ogunniyi**, Austin RJ Downey, Subramani Sockalingam, Han Liu, and Simon Laflamme. In situ monitoring of impact-induced capacity loss in structural batteries with flexible electronics. *Journal of Power Sources*, 659:238417, 2025
- [6] Yanzhou Fu, Austin RJ Downey, Lang Yuan, Hung-Tien Huang, and **Emmanuel A. Ogunniyi**. Simulation-in-the-loop additive manufacturing for real-time structural validation and digital twin development. *Additive Manufacturing*, 98:104631, 2025

- [5] **Emmanuel A. Ogunniyi**, Han Liu, Austin RJ Downey, Simon Laflamme, Jian Li, Caroline Bennett, William Collins, Hongki Jo, and Paul Ziehl. In situ assembly enabling adhesive-free bonding of large area electronic sensors to concrete for structural health monitoring. *Smart Materials and Structures*, 33(10):105047, 2024
- [4] **Emmanuel A. Ogunniyi**, Claire Drnek, Seong Hyeon Hong, Austin R.J. Downey, Yi Wang, Jason D. Bakos, Peter Avitabile, and Jacob Dodson. Real-time structural model updating using local eigenvalue modification procedure for applications in high-rate dynamic events. *Mechanical Systems and Signal Processing*, 195:110318, jul 2023. doi:10.1016/j.ymssp.2023.110318
- [3] **Emmanuel A. Ogunniyi**, Alexander Vereen, Austin R.J. Downey, Simon Laflamme, Jian Li, Caroline R Bennett, William Collins, Hongki Jo, Alexander Henderson, and Paul Ziehl. Investigation of electrically isolated capacitive sensing skins on concrete to reduce structure/sensor capacitive coupling. *Measurement Science and Technology*, feb 2023. doi:10.1088/1361-6501/acbb97
- [2] John Adeniyi., **Emmanuel A. Ogunniyi.**, and Ifeanyi Okafor. Biodiesel production from palm oil using heterogeneous base catalyst. *International Journal of Scientific Research in Science, Engineering and Technology*, pages 80–90, 11 2019. doi:10.32628/ijrsrset196615
- [1] Oluwaseun Adekoya., Oluwatimilehin Oluwajire., and **Emmanuel A. Ogunniyi.** Design, modeling and fabrication of a portable solar powered suction pump for use in industries and autoclinics. *International Journal of Scientific Research in Science, Engineering and Technology*, pages 287–297, 10 2019. doi:10.32628/ijrsrset1196559

Conference Proceedings

- [10] Killedar Digambar, **Emmanuel A. Ogunniyi**, Fu Yanzhou, Downey Austin, Yuan Lang, and D Folsom Matthew. In-situ monitoring of geometric drift and thermo-mechanical modeling toward a digital twin for thin-wall fused filament fabrication. In *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2026*. SPIE, 2026
- [9] **Emmanuel A. Ogunniyi**, Han Liu, Austin RJ Downey, Simon Laflamme, and Li. Impact monitoring of embedded batteries in sandwich composites with integrated soft elastomeric capacitors. In *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2025*, volume 12949, pages 267–274. SPIE, 2025
- [8] **Emmanuel A. Ogunniyi**, Han Liu, John White, Austin RJ Downey, Simon Laflamme, Jian Li, Caroline Bennett, William Collins, Hongki Jo, and Paul Ziehl. Performance evaluation of flexible capacitive sensors on non-uniform surfaces. In *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2024*, volume 12949, pages 267–274. SPIE, 2024
- [7] **Emmanuel A. Ogunniyi**, Joud N Satme, and Austin RJ Downey Jr. Online model-based structural damage detection in electronic assemblies. In *Active and Passive Smart Structures and Integrated Systems XVIII*, volume 12946, pages 283–287. SPIE, 2024
- [6] Goerge Anthony, Connor Madden, **Emmanuel A. Ogunniyi**, Austin RJ Downey, Ryan Limbaugh, Jarret Peskar, Jingjing Bao, and Xinyu Huang. Exploratory investigation

- of early detection for high-c discharge-induced failure in 18650 lithium-ion batteries. In *Health Monitoring of Structural and Biological Systems XVIII*, volume 12951, pages 135–139. SPIE, 2024
- [5] **Emmanuel A. Ogunniyi**, Alexander B Vereen, and Austin RJ Downey. Microsecond model updating for 2d structural systems using the local eigenvalue modification procedure. *International Workshop on Structural Health Monitoring*, 2023
 - [4] Alexander B Vereen, **Emmanuel A. Ogunniyi**, Austin RJ Downey, Erik Blasch, Jason D Bakos, and Jacob Dodson. Optimal sampling methodologies for high-rate structural twinning. In *2023 26th International Conference on Information Fusion (FUSION)*, pages 1–8. IEEE, 2023
 - [3] **Emmanuel A. Ogunniyi**, Han Liu, Austin RJ Downey, Simon Laflamme, Jian Li, Caroline Bennett, William Collins, Hongki Jo, and Paul Ziehl. Soft elastomeric capacitors with an extended polymer matrix for strain sensing on concrete. In *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2023*, volume 12486, pages 262–270. SPIE, 2023
 - [2] Alexander B Vereen, **Emmanuel A. Ogunniyi**, Austin RJ Downey, Jacob Dodson, Adriane G Moura, and Jason D Bakos. Online implementation of the local eigenvalue modification procedure for high-rate model assimilation. In *Society for Experimental Mechanics Annual Conference and Exposition*, pages 121–127. Springer, 2023
 - [1] **Emmanuel A. Ogunniyi**, Austin R. J. D., and Jason B. Development of a real-time solver for the local eigenvalue modification procedure. page 51. SPIE, 4 2022. doi:10.1117/12.2613208

Conference Presentation

- [4] **Emmanuel A. Ogunniyi**, White John, Han Liu, Austin RJ Downey, Simon Laflamme, Jian Li, Caroline Bennett, William Collins, Hongki Jo, and Paul Ziehl. Performance evaluation of flexible capacitive sensors on non-uniform surfaces. In Daniele Zonta, Zhongqing Su, and Branko Glisic, editors, *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2024*, page 51. SPIE, 4 2024
- [3] **Emmanuel A. Ogunniyi**, Joud Satme, and Austin RJ Downey. Online model-based structural damage detection in electronic assemblies. In Daniele Zonta, Zhongqing Su, and Branko Glisic, editors, *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2024*, page 51. SPIE, 4 2024
- [2] Alexander B Vereen, **Emmanuel A. Ogunniyi**, Austin RJ Downey, Erik Blasch, Jason D Bakos, and Jacob Dodson. Optimal sampling methodologies for high-rate structural twinning. In *2023 26th International Conference on Information Fusion (FUSION)*, pages 1–8. IEEE, 2023
- [1] **Emmanuel A. Ogunniyi**, Austin R. J. D., and Jason B. Development of a real-time solver for the local eigenvalue modification procedure. In Daniele Zonta, Zhongqing Su, and Branko Glisic, editors, *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2022*, page 51. SPIE, 4 2022. doi:10.1117/12.2613208

Invited Presentations

- [2] **Emmanuel A. Ogunniyi**, Alexander B Vereen, and Austin RJ Downey. Microsecond model updating for 2d structural systems using the local eigenvalue modification procedure. In *High-rate Structural Monitoring, Damage Detection, Prognostics, and Reactions Working Group*, September 05, 2023
- [1] **Emmanuel A. Ogunniyi**. Reduced order model-based framework for microsecond model updating of two-dimensional structural systems using the local eigenvalue modification procedure. In *High-rate Structural Monitoring, Damage Detection, Prognostics, and Reactions Working Group*, April 15, 2025

Awards and Honors

- [13] **International Society of Information Fusion**. Awarded up to \$1000 to attend international conference of fusion at venice, italy, 2024
- [12] **SPIE Smart Structures + NDE**. Awarded \$700 to present a paper at spie smart structures + nde conference at longbeach, califonia, 2024
- [11] **University of South Carolina**. Awarded \$2000 to present a paper at spie smart structures + nde, 2024
- [10] **University of South Carolina**. Awarded \$1000 to present a paper at international conference of fusion, 2023
- [9] **University of South Carolina**. Awarded \$1200 to present a paper at spie smart structures + nde, 2022
- [8] **Ph.D full-time funding**. Graduate funding for full-time ph.d in mechanical engineering for five years at the university of south carolina, columbia, 2020
- [7] **Pan African Scholarship**. Full-time masters degree in mechanical engineering at pan african university institute for basic sciences, technology and innovation (pausti)/jomo kenyatta university of agriculture and technology, awarded to top 1% citizens of african countries by the african union, 2019
- [6] **African Railway Center of Excellence (ARCE)**. Scholarship for full-time masters degree in railway engineering (mechanical options) at addis ababa institute of technology, ethiopia, award to 30 african citizens, 2019
- [5] **Federal Government of Nigeria Scholarship Board Award**. Top performing student in nigeria institution, maintaining a cgpa of above 4.0/5.0 every session, 2017-2018
- [4] **NNPC/TOTAL Merit Award Scholarship for Undergraduates**. Outstanding performing undergraduate for 5years, 2015-2018
- [3] **Agbami Medical/Engineering Scholarship Award (AMES)**. Award for outstanding performing undergraduate through selection exams and with continuous achievement of cgpa of 3.5/5.0 every session, 2015-2018
- [2] **MTN Foundation Scholarship**. Outstanding 500 science student in the nigeria, awarded to top 1% science student in nigeria., 2015-2018

- [1] **Dean's list.** On dean's list for 8 out of 10 semesters at federal university of tech. akure., 2014 - 2018

Courses Taught

- [5] **EMCH-368 Mechatronics.** Fall 2023
- [4] **EMCH-368 Mechatronics Lab.** Fall 2023
- [3] **EMCH-368 Mechatronics.** Spring 2022
- [2] **EMCH-368 Mechatronics Lab.** Spring 2022
- [1] **MEE 210 Mechanical Engineering design.** Fall 2019

Grants and Contracts

- This are grants and contracts contributed to

- [12] **National Science Foundation** (awaiting decision). "TRACK II FEC: Physics-Based Digital Twin for Urban Flood Infrastructure Operations and Planning", June 2026 - May 2030, \$4,140,000, Serving as Co-PI
- [11] **Benedict College.** "B.E.S.T Innovation: Development of Low-cost Robotic Arm Additive Manufacturing system for Advanced Situ Monitoring Research and Education", March 2026 - June 2026, \$2,000, Serving as PI
- [10] **South Carolina** (awaiting decision). "SC EPSCoR: Low-Power Biomedical AI using FPGA-Optimized Neural Networks for Real-Time Health Monitoring", May 2026 - Nov 2027, \$70,000, Serving as Co-PI
- [9] **National Science Foundation.** "Excellence in Research: Real-Time Monitoring and Structural Validation for Continuous Fiber Printing", Oct 2025 - Sept 2028, \$887,740, Serving as PI
- [8] **National Science Foundation.** "Collaborative Research: SHF: Small: Sub-millisecond Topological Feature Extractor for High-Rate Machine Learning", May 2023 - May 2025, \$250,000
- [7] **National Science Foundation.** "CAREER: Data-Driven Control of High-Rate Dynamic Systems", February 2023 - January 2028, \$551,876
- [6] **Air Force Office of Scientific Research.** "2021 YIP: Real-time Model Updating for Structures in Shock Environments", May 2021 - May 2024, \$450,000
- [5] **National Science Foundation.** "Collaborative Research:SHF:Medium:Machine Learning on the Edge for Real-Time Microsecond State Estimation of High-Rate Dynamic Events", August 2020 - July 2024, \$706,248
- [4] **United States Department of Transportation.** "Robust wireless skin sensor networks for long-term fatigue crack monitoring of bridges", May 2020 - May 2023, \$80,000
- [3] **Air Force Research Laboratory.** "Subcontract on Prime Contract No FA8651-16-D- 0311, Task Order 0004. Real Time High-rate Decision Making for Functional Prognosis of Complex Mechanical Systems", April 2019 - March 2021, \$39,999
- [2] **University of South Carolina.** "ASPIRE-AI: Real-Time Surrogate Model Updating for Structures Experiencing High-Rate Dynamics", August 2019 - July 2021, \$15,000, PI: Austin Downey; Co-PI: Yi Wang
- [1] **National Science Foundation.** "CRII: Algorithms and Methodologies for Real-Time Decision-Making of Mission-Critical Structures Experiencing High-Rate Dynamics", March 2019 - February 2021, \$191,000

M.Sc. Thesis

- [1] **Emmanuel A. Ogunniyi.** Elastic Sensing Skin for Monitoring of Concrete Structures. MSc. Thesis, University of South Carolina, Columbia, 2023

Ph.D. Dissertation

- [1] **Emmanuel A. Ogunniyi.** Reduced Order Model-based Framework for Microsecond Model Updating of Two-Dimensional Structural Systems Using the Local Eigenvalue Modification Procedure. Ph.D Dissertation, University of South Carolina, Columbia, 2025

Professional affiliations

- [6] The National Society of Black Engineers (NSBE), 2023 - Present
- [5] Society for Experiment Mechanics (SEM), 2023 - Present
- [4] The Institute of Electrical and Electronics Engineers (IEEE), 2023 - Present
- [3] The American society of Mechanical Engineers (ASME), 2022 - Present
- [2] The international society for optics and photonics (SPIE), from 2022. Present
- [1] The Nigerian Society of Engineers (NSE), 2018 - Present

Organized Trainings/Sessions

- [1] Quanser robotics training. Cordinated Robotics Training at Benedict College Office of Research which includes trainings Robotics Arms, Quanser Bots, Autonomous cars

Selected Mentorship at Benedict College

- [3] Taleah brown; computer science. “High-rate Machine learning algorithm for defect dtection in FDM”, 2026 - Present
- [2] Kaleab alemu; computer science. “Small-scale robotic-arms cordination for manufacturing assembly task”, 2026 - Present
- [1] Jalil sweat; computer science. “In situ monitoring techniques for additively manufactured components”, 2025

Selected Mentorship at University of South Carolina as Ph.D student

- [7] Mark zheng; mechanical engineering. “Image-based Tracking of UAV Systems in Flight”, 2024 - present. Awarded: 2024 McNair Summer Fellowship \$3,000
- [6] Ryan limbaugh; mechanical engineering. “Battery Degradation Modeling”, 2023 - present
- [5] Connor madden; mechanical engineering. “Electric vertical lift system testbed development”, 2022 - 2024. Transitioned to a M.S. at the USC
- [4] Corinne smith; mechanical engineering,. “UAV Deployed sensors for hydrologic parameter sensing”, 2021. Awarded: Magellan Scholar, \$3,000; 2022 McNair Summer Fellowship \$2,500; Mike and Ann Sutton Fellowship \$1500; Magellan Voyager program (travel grant) \$500; Honors College Research Grant Award \$1,500; AIAA Guidance, Navigation, and Control Technical Committee - Undergraduate Conference Experience \$600 travel + conference registration. Awarded NSF GRFP and attend Carnegie Mellon University for a Ph.D
- [3] John white; mechanical engineering. “Signal Conditioning Electronics for Low-cost Nuclear Magnetic Resonance”, 2021 - 2024. Awarded: Magellan Scholar, \$3,000
- [2] Chris nelson; mechanical engineering. “Sensing Skins for Structural Health Monitoring”, 2022 - 2022

- [1] William bowers; mechanical engineering. “Large Area Sensing Skins for Crack Detection”, 2021

Academic Reviewer (select and current)

Journal name	Publisher
Measurements	Elsevier
Batteries	MDPI
Sensors	MDPI
Engineering Research Express	IOPScience
Applied Sciences	MDPI
Machines	MDPI
Eng	MDPI
Smart Materials and Structures	IOPScience
Energies	MDPI
Flexible and Printed Electronics	IOPScience

Volunteering/Community service

- [9] DiscoverUofSC. Reviewer for undergraduate sessions for chemistry, physics, and math, engineering and computing, 2024
- [8] South Carolina high school science fair. Served as project judge for all class range, 2024
- [7] Homeless shelter. Volunteering activities at oliver gospel mission, the homeless shelter, 2022
- [6] DiscoverUofSC. Reviewer for undergraduate sessions for chemistry, physics, and math, engineering and computing, 2022
- [5] President NYSC SDGS. Community development service, 2019
- [4] Heal for Africa Initiative. World largest health campaign community, 2019
- [3] Get Educated Get a Skill. Facilitator –a non-governmental organization that reaches out to young students by promoting quality education and skill acquisition., 2019
- [2] The Orphanage Empowerment Society. Program photographer and skill trainee and orphanage outreaches, 2018-2019
- [1] FUTA Academic Transformation team. Encouraging and mentoring student on how to become an academic giant and how to get scholarships as an undergraduate, 2016-2018

References

- Austin Downey, Mechanical Engineering University of South Carolina, Columbia, SC, email: austindowney@sc.edu
- Jason Bakos, Computer Science and Engineering, University of South Carolina, Columbia, SC, email: jbakos@cse.sc.edu
- Simon Laflamme, Civil and Environmental Engineering, Iowa State University, Ames, Iowa, email: laflamme@iastate.edu